

The Message from the Nucleus

II PUC · Ch 5 · Molecular Basis of Inheritance

Starring DNA the Librarian 📜 & mRNA the Messenger ✉️

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A Secret Instruction

Deep inside a stomach cell, DNA the Librarian guarded a secret instruction: "Make an enzyme to digest food." But DNA can never leave the **nucleus**. "I will send a messenger," it said. This flow of information — DNA to RNA to protein — is the **central dogma**, proposed by **Francis Crick**. "Let's follow the message!"

💡 **REMEMBER**

Central dogma: DNA → (transcription) → RNA → (translation) → protein. In some viruses RNA → DNA by reverse transcription.



The Twisted Ladder

DNA was a **double helix**, described by **Watson and Crick** in 1953. Its two strands run in opposite directions — **antiparallel**. "My base-pairing rules never change," said DNA. "**Adenine** pairs with **thymine** using two hydrogen bonds; **guanine** pairs with **cytosine** using three." One full turn of the helix is 3.4 nm long and has about 10 base pairs.

💡 **REMEMBER**

Chargaff's rule: in double-stranded DNA, A = T and G = C, so purines = pyrimidines.



Copying the Message

The enzyme **RNA polymerase** arrived at a **promoter** and unzipped a short stretch of DNA. Reading the **template strand** (3′ → 5′), it built an RNA copy that matched the **coding strand** — except that **uracil** replaced thymine. At the **terminator** it stopped. This copying of DNA into RNA is called **transcription**.

💡 **REMEMBER**

A transcription unit has a promoter, a structural gene and a terminator.



Editing the Draft

In eukaryotes the first copy, **hnRNA**, is a rough draft. "Cut out the non-coding **introns** and join the coding **exons**," ordered the editor — this is **splicing**. Then a methyl guanosine triphosphate cap was added at the 5′ end — **capping** — and a long chain of adenylate residues at the 3′ end — **tailing**. Now the **mRNA** was ready to travel.

> 💡 **REMEMBER**
>
> **Bacteria have no splicing; their mRNA can be translated even while it is being transcribed.**



Journey to the Ribosome

mRNA slipped out through a **nuclear pore** and reached the protein factory, the **ribosome**. The message was written in three-letter words called **codons** — the **genetic code**. Of 64 codons, 61 code for amino acids. **AUG** means "start" (methionine); UAA, UAG and UGA mean "stop". The code is **degenerate** — many codons can name one amino acid — and nearly **universal**.

💡 **REMEMBER**

George Gamow suggested a triplet code; Nirenberg and Khorana helped decode it.



Building the Protein

Delivery trucks called **tRNA** brought amino acids. Each tRNA had an **anticodon** that matched a codon on mRNA. First the amino acids were loaded — **charging of tRNA** (aminoacylation). Then came **translation**: initiation at AUG, elongation as **peptide bonds** formed, and termination when a release factor met a stop codon.

💡 **REMEMBER**

In bacteria, 23S rRNA acts as a ribozyme that catalyses peptide bond formation.



Switching Genes On and Off

"A cell does not make every protein all the time," said DNA. In bacteria, **Jacob and Monod** discovered the **lac operon**. A **repressor** normally blocks the operator. When lactose arrives it acts as an **inducer**, removes the repressor and switches on the genes for digesting lactose. This is **gene regulation**.

💡 **REMEMBER**

The lac operon has the structural genes z (β-galactosidase), y (permease) and a (transacetylase).



The Enzyme Is Ready

The new chain of amino acids folded into a working enzyme and set off to digest food. "The message reached the factory!" cheered mRNA. DNA smiled from its library. "Remember: the order of bases in DNA decides the order of amino acids in a protein — and the protein decides the trait."

💡 **REMEMBER**

The Human Genome Project found about 3×10^9 base pairs and about 20,000 genes in humans.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

nucleus p.1 · central dogma p.1 · Francis Crick p.1 · double helix p.2 · Watson and Crick p.2 · antiparallel p.2 · Adenine p.2 · thymine p.2 · guanine p.2 · cytosine p.2 · RNA polymerase p.3 · promoter p.3 · template strand p.3 · coding strand p.3 · uracil p.3 · terminator p.3 · transcription p.3 · hnRNA p.4 · introns p.4 · exons p.4 · splicing p.4 · capping p.4 · tailing p.4 · mRNA p.4 · nuclear pore p.5 · ribosome p.5 · codons p.5 · genetic code p.5 · AUG p.5 · degenerate p.5 · universal p.5 · tRNA p.6 · anticodon p.6 · charging of tRNA p.6 · translation p.6 · peptide bonds p.6 · Jacob and Monod p.7 · lac operon p.7 · repressor p.7 · inducer p.7 · gene regulation p.7



QUICK QUIZ

Do you remember?

1. Which molecule carries the genetic message from the nucleus to the ribosome?

- tRNA
- mRNA
- rRNA
- DNA

2. What are the base-pairing rules in DNA?

- A–G and T–C
- A–T and G–C
- A–U and G–C
- A–C and G–T

3. What is the function of ribosomes?

- Making lipids
- Protein synthesis
- Storing water
- Cell division

4. The start codon is:

- UAA
- AUG
- UGA
- UAG

5. Splicing removes the:

- Exons
- Introns
- Cap
- Poly-A tail

Answers: 1-mRNA · 2-A–T and G–C · 3-Protein synthesis · 4-AUG · 5-Introns